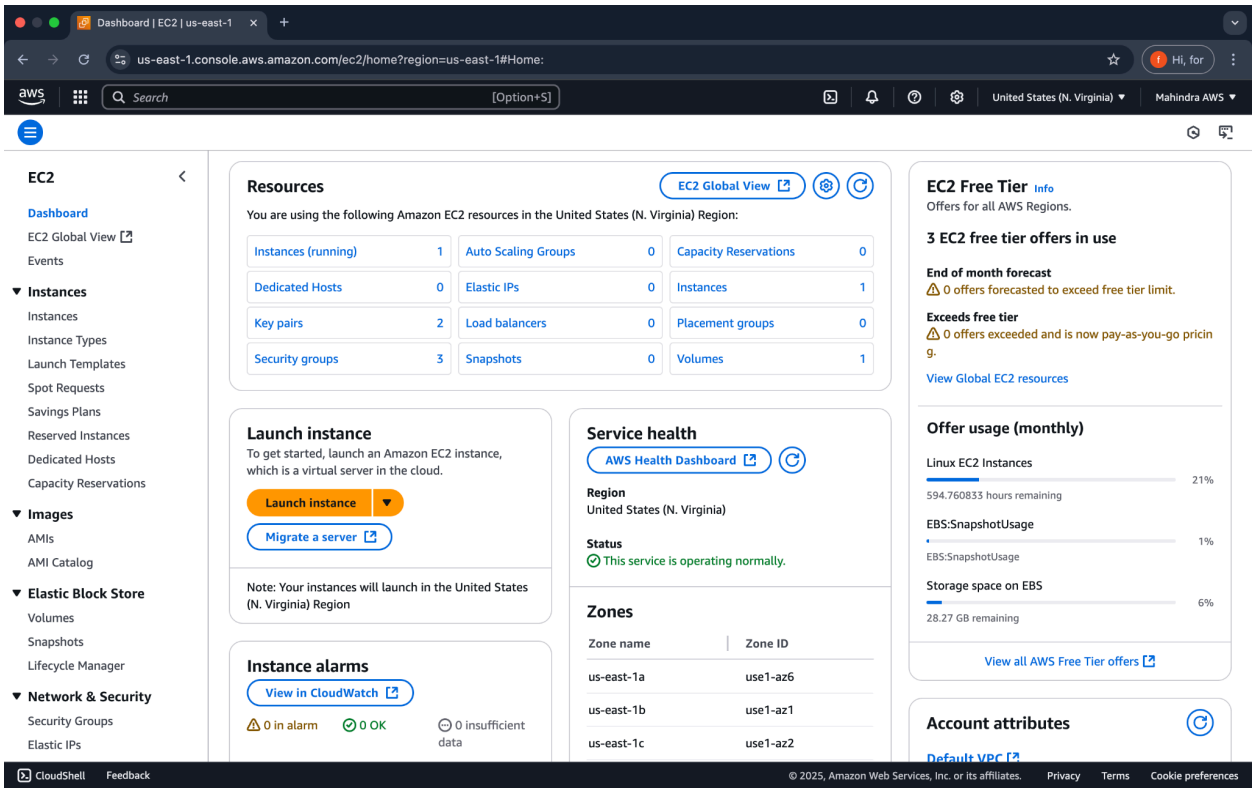
High-Performing Architectures

Author : Mahindra Guptha Kotha
Mail ID: mkotha@smu.edu

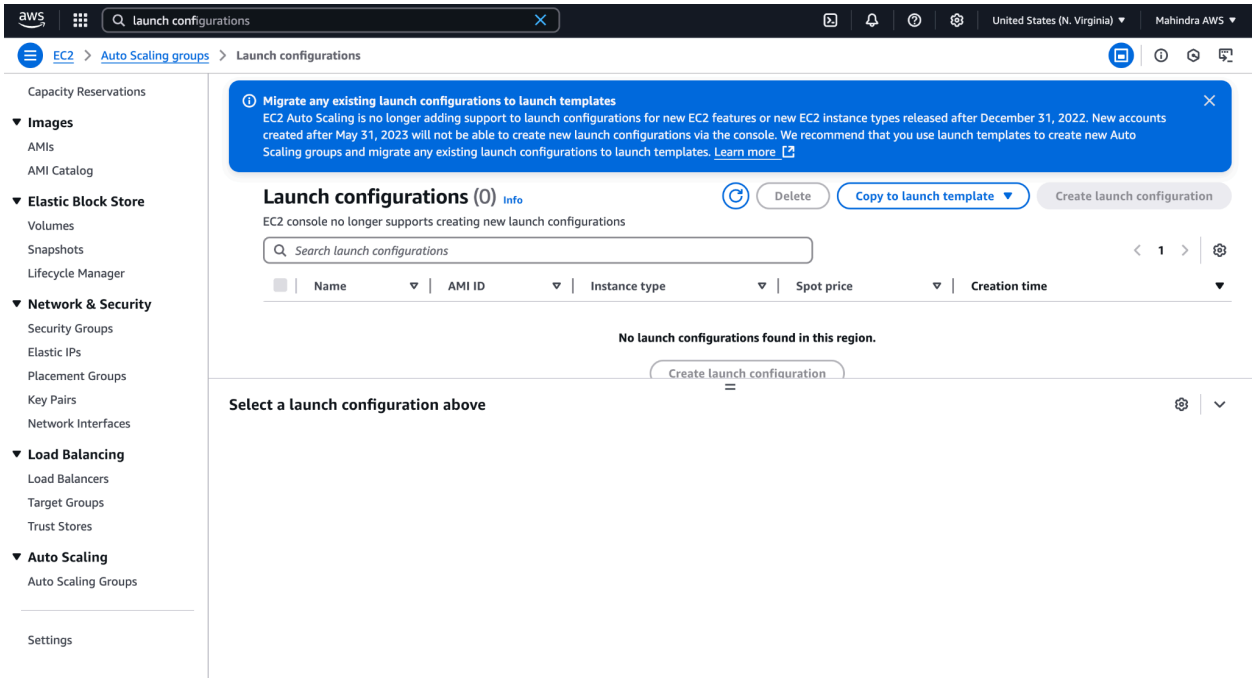
EXERCISE 11.1

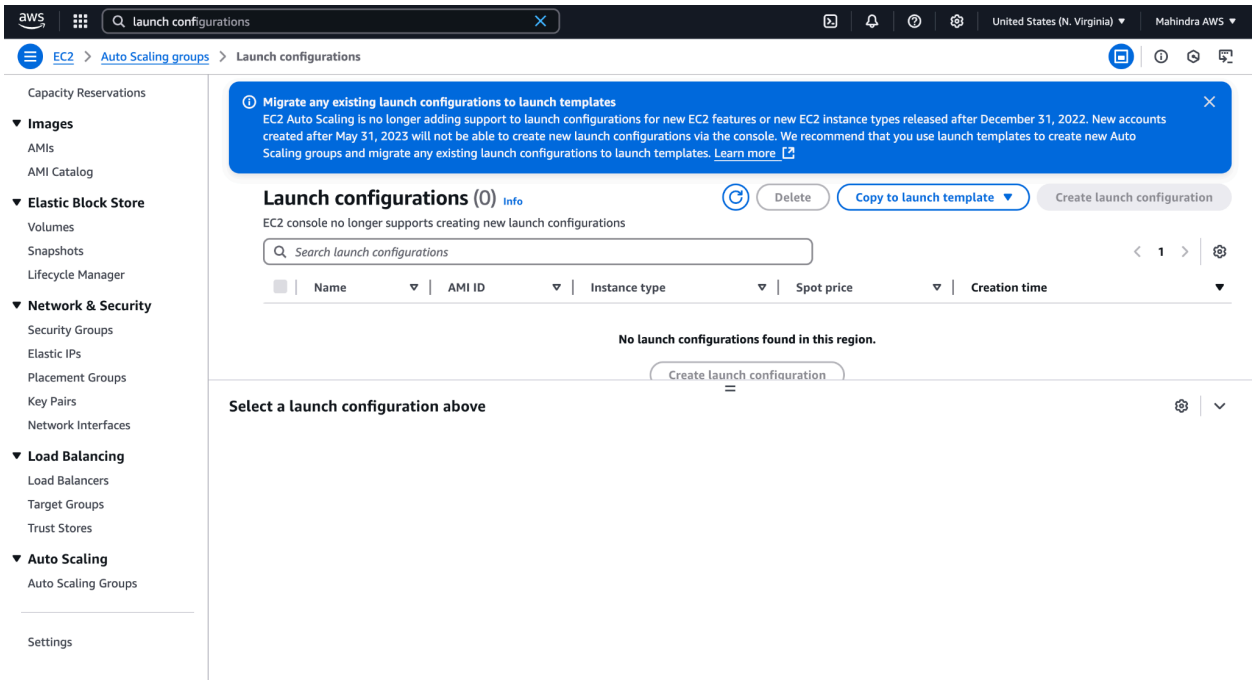
Configure and Launch an Application Using Auto Scaling

1. From the EC2 Dashboard, create a launch configuration using the Ubuntu Server LTS AMI on the Quick Start tab and a t2.micro instance type.



2. On the Create Launch Configuration page, name your configuration. You don't need to select an IAM role or enable monitoring for this exercise.

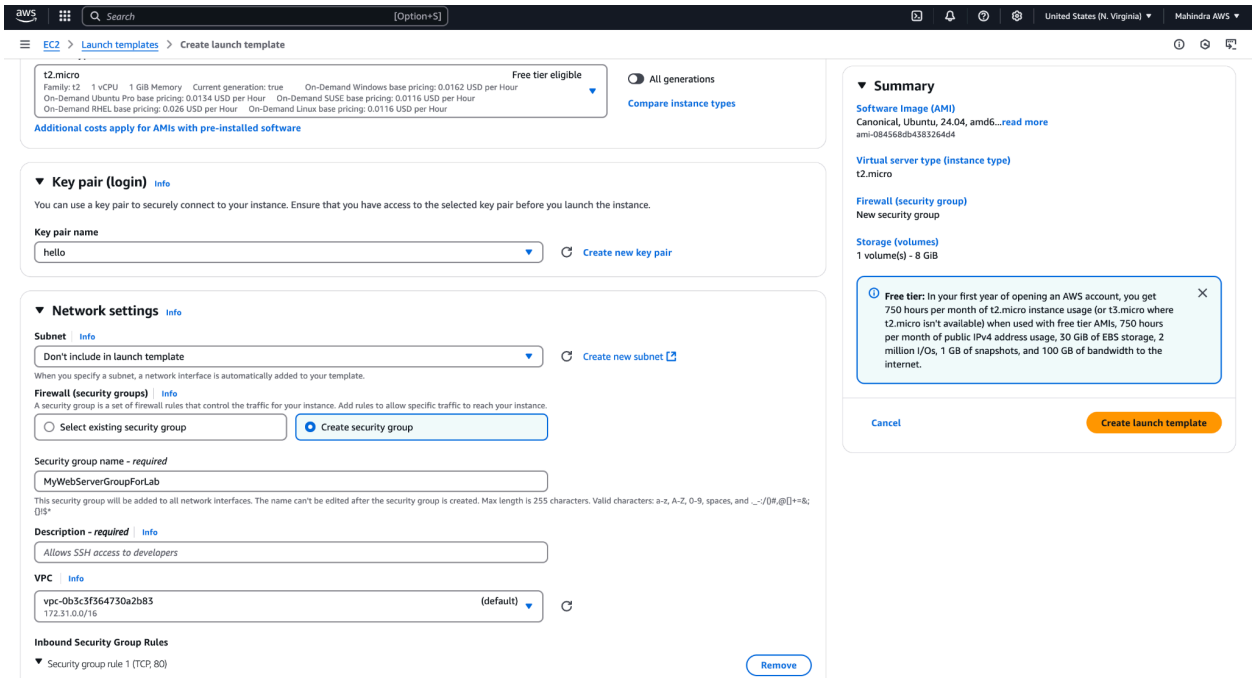

launch configurations
United States (N. Virginia)
Mahindra AWS
EC2 > Auto Scaling groups > Launch configurations
Capacity Reservations
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
Migrate any existing launch configurations to launch templates
EC2 Auto Scaling is no longer adding support to launch configurations for new EC2 features or new EC2 instance types released after December 31, 2022. New accounts created after May 31, 2023 will not be able to create new launch configurations via the console. We recommend that you use launch templates to create new Auto Scaling groups and migrate any existing launch configurations to launch templates. Learn more
Launch configurations (0) Info
Delete
Copy to launch template
Create launch configuration
EC2 console no longer supports creating new launch configurations
Search launch configurations
Name
AMI ID
Instance type
Spot price
Creation time
No launch configurations found in this region.
Create launch configuration
Select a launch configuration above

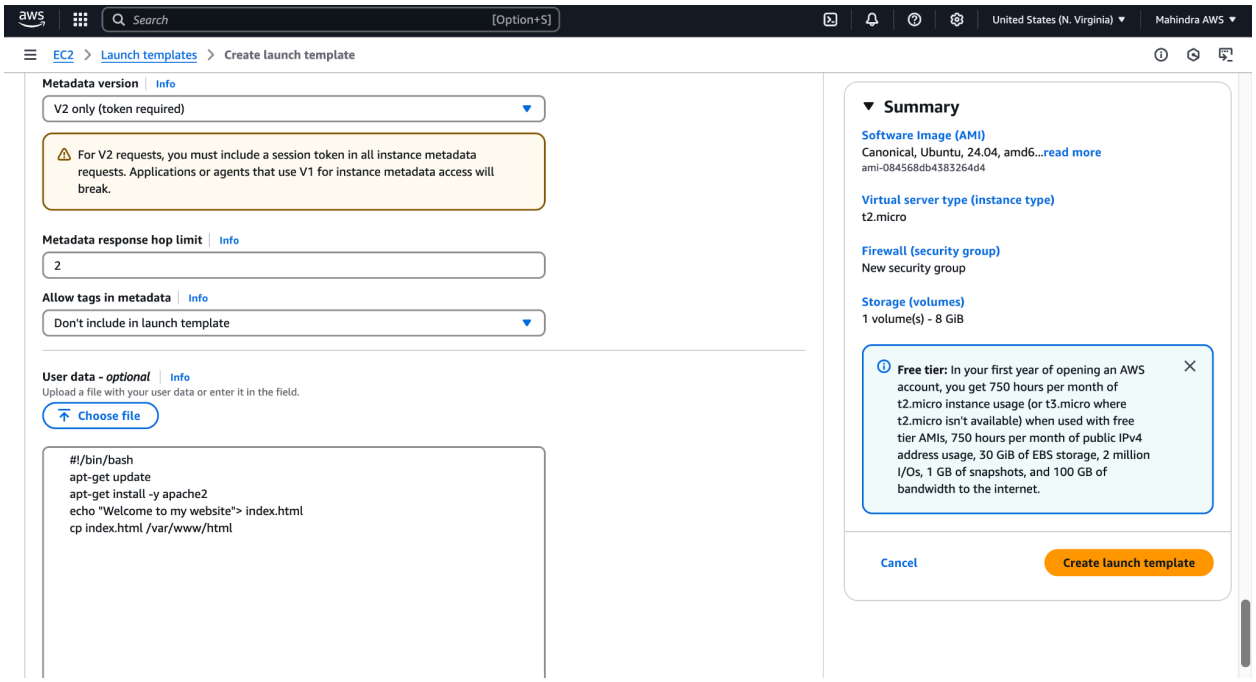

launch configurations
United States (N. Virginia)
Mahindra AWS
EC2 > Auto Scaling groups > Launch configurations
Capacity Reservations
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
Migrate any existing launch configurations to launch templates
EC2 Auto Scaling is no longer adding support to launch configurations for new EC2 features or new EC2 instance types released after December 31, 2022. New accounts created after May 31, 2023 will not be able to create new launch configurations via the console. We recommend that you use launch templates to create new Auto Scaling groups and migrate any existing launch configurations to launch templates. Learn more
Launch configurations (0) Info
Delete
Copy to launch template
Create launch configuration
EC2 console no longer supports creating new launch configurations
Search launch configurations
Name
AMI ID
Instance type
Spot price
Creation time
No launch configurations found in this region.
Create launch configuration
Select a launch configuration above

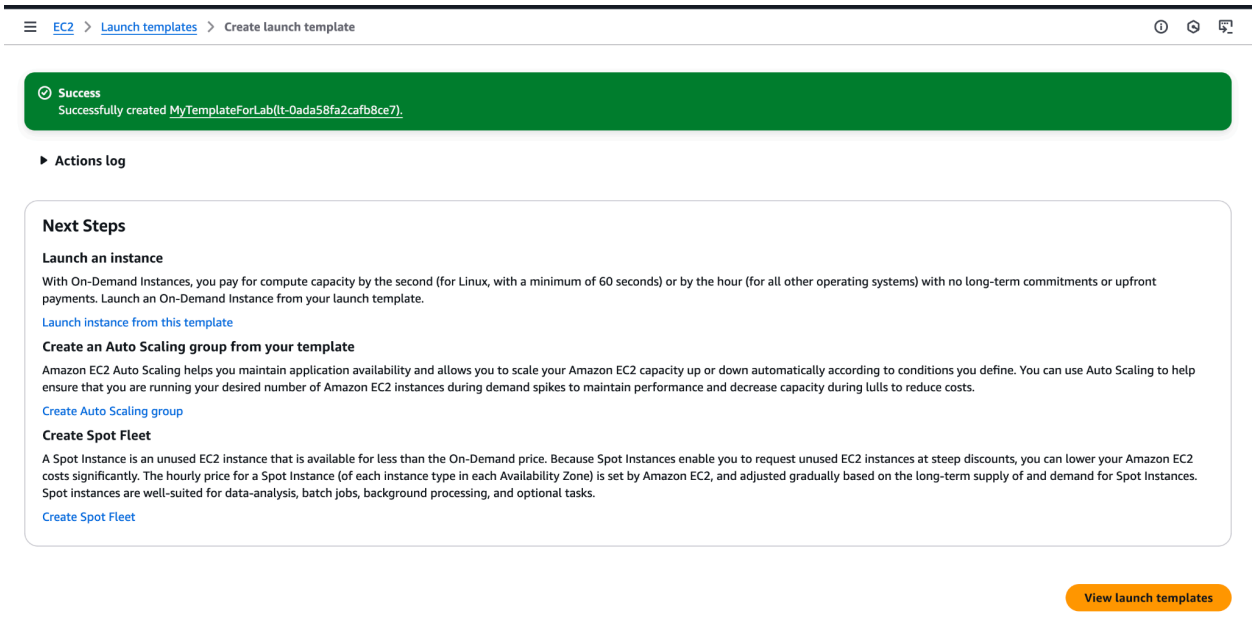
Save the following commands to a file on your local computer, name the file `start.sh`, expand the Advanced Details section, click the As File radio button, and select your `start.sh` script from your computer. This will install the Apache web server and create a simple `index.html` web page:

```
    #!/bin/bash
    apt-get update
    apt-get install -y apache2
    echo "Welcome to my website"> index.html
```

3. `    cp index.html /var/www/html`
4. Choose (or create) a security group that will permit all HTTP traffic via port 80 and then create the configuration, making sure to select a valid key pair so that you'll be able to use SSH to log into the instance later if necessary.







5. Create an Auto Scaling group that will use your new launch configuration. Give your group a name and leave the Group Size value at 1 instance.

6. Select a virtual private cloud (VPC) into which your instances will launch, click once inside the Subnet field, and select the subnet where you'd like your instances to live. Assuming you're working with your default VPC, any one of the public subnets you'll be shown should be fine.
7. On the Configure Scaling Policies page, set Minimum Capacity to 1 and Maximum Capacity to 2.
8. Select Target Tracking Scaling Policy and then leave the Metric Type value as Average CPU Utilization and edit its Target Value to 5 percent.
   This unusually low value will make it easier to confirm that the autoscaling is working—normally a value of 70 percent to 80 percent would make sense.
9. For this exercise, you don't need to configure notifications or tags. Click Create to finish creating the group.
10. Your group will immediately try to start the first instance. It may take a few minutes for everything to load. When it does, point your browser to the IP address associated with the instance (you can retrieve that from the EC2 Instances Dashboard).
    Considering how small the `index.html` page is, it might be hard to get the Auto Scaling tool to launch a second instance. You'll probably need to "cheat" by running a busywork command such as the following on the server command line to raise your CPU level high enough:

    ```
    $ while true; do true; done
    ```

EXERCISE 11.2

Sync Two S3 Buckets as Cross-Region Replicas

1. Use the S3 Console to create two buckets in two separate regions. One will be your source and the other your destination.
2. Click the name of your source bucket in the S3 Console, then click the Management tab, then the Create Replication Rule button. If versioning wasn't already enabled for the bucket, you can click the Enable Versioning button to set it.
3. Define the objects you want replicated from your source bucket. Choose Apply To All Objects for this exercise.
4. In the Destination Definition section, select the name of the second bucket you created as the destination.
5. Choose From Existing IAM Roles and select Create New Role from the drop-down menu. This will create the permissions necessary to move assets between buckets. You can then review your settings and save the rule.
6. Save your rule and confirm the replication is working by uploading a file to the source bucket and then checking to see that a copy has been created in the destination bucket.

EXERCISE 11.3

Upload to an S3 Bucket Using Transfer Acceleration

Enable S3 Transfer Acceleration on an existing bucket (substitute the name of your bucket for *my-bucket-name*).

```
$ aws s3api put-bucket-accelerate-configuration \

    --bucket my-bucket-name \

    --accelerate-configuration Status=Enabled
```

Transfer a file by specifying the `s3-accelerate` endpoint. Make sure you're using the correct file and bucket names and region.

```
$ aws s3 cp filename.mp4 s3://my-bucket-name \

    --region us-east-1 \

    --endpoint-url
http://s3-accelerate.amazonaws.com
```

Create and Deploy an EC2 Load Balancer

1. Create one instance in each of two subnets. You select the subnets on the Configure Instance Details page. Make a note of the subnet designations you choose.
2. Use SSH to access your instances, install the Apache web server, and create unique `index.html` files containing a short note describing the server—perhaps including

the server's IP address. This will make it easier to know which instance your load balancer has loaded when you test everything out later.

3. From the EC2 Dashboard, create a load balancer. Select an application load balancer, give it a name, and specify the “Internet-facing” scheme, IPv4 for the IP address type, and the default HTTP and port 80 values for the listener. Make sure you're using the right VPC (the default should be fine) and select (and note) two availability zones. For this exercise, you can ignore the warning about using an unencrypted listener.
4. Select (or create) a security group that allows access to port 80.
5. Click the Create Target Group link and change the Target Type value to IP and add a path to a file the health check can use. For this exercise, enter `/index.html`.

6. On the Register Targets page, register your two EC2 instances as targets to the group using their private IPs. You can retrieve those private IP addresses from the EC2 Instances console or from the command prompt of the instance itself using the `ip addr` command.
7. Once your balancer is active (which can take a couple of minutes), get the DNS name of your balancer from the Load Balancers console and paste it into your browser. The `index.html` page of one of your instances should load. Refresh the browser a few times so that you can confirm that both instances are being accessed.
8. If you like, you can shut down one of the instances without causing any interruption to the load balancer service.

EXERCISE 11.5

Create a Nested Stack

In this exercise, you'll create a nested stack that creates an EC2 Auto Scaling group and the supporting network infrastructure. To complete this exercise, you'll need to know the name of a Secure Shell (SSH) key pair in the region you're using.

1. Visit http://awscsa.github.io and click Chapter 14. Download the `web-stack.json` and `network-stack.json` CloudFormation templates.
2. Create an S3 bucket and upload the `network-stack.json` template to the bucket.
3. Edit the `web-stack.json` template and locate the `Network-Stack` resource. Change the `TemplateURL` value to the URL of the `network-stack.json` template in S3.
4. Upload the `web-stack.json` template to the same S3 bucket you used in step 2.
5. Run the following AWS CLI command:

```
aws cloudformation create-stack --stack-name web-stack

    --template-url [URL of the web-stack.json template]

    --parameters ParameterKey=KeyName,ParameterValue=[the
name of your SSH keypair]
```

6. `    --capabilities CAPABILITY_NAMED_IAM`
7. CloudFormation will begin to create the `Web` stack and the nested network stack. Enter the following command to view the stacks:
8. `aws cloudformation describe-stacks`

9. To delete the stacks, run the following command:

```
aws cloudformation delete-stack --stack-name web-s
```

EXERCISE 11.6

Create a CloudWatch Dashboard

1. So that you'll get a better idea of how things will look, make sure you have at least one EC2 instance running and at least one S3 bucket with some files in it.
2. From the CloudWatch console, click Dashboards, click Create Dashboard, and give your dashboard a name. On the Add Widget page, select the Line metric. Then select the Metrics option.
3. From the Metrics displayed in the bottom half of the screen, select the appropriate AWS region and click EC2 and then click Per-Instance Metrics. Check the boxes for a few metrics associated with your running instance. StatusCheckFailed, NetworkOut, NetworkIn, and CPUUtilization would make for a nice sample. Note how those metrics are now represented in the graph at the top of the screen. When you're satisfied with your selections, click Create Widget.
4. Your first widget will now appear in your dashboard. You can click the widget to display it full-screen. Selecting a particular metric from the legend at the bottom of

the graph will show you only that metric, making it easier to understand what the values represent.

5. If your widget is still full-screen, click Close to return to the dashboard.

6. Click Add Widget to create a second widget. This time, select Stacked Area and then Storage Metrics. Select S3 metrics and then the NumberOfObjects and BucketSizeBytes metrics for your bucket. Create the widget.

7. Once you've added all the widgets and metrics you'll need to monitor, click Save Dashboard.

   From now on, whenever you browse to CloudWatch, you'll be able to retrieve and view this dashboard with all of its metrics. Bear in mind that, outside of the Free Tier, a single dashboard will cost around $3 per month.